

MICHAEL CAI

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PROFESSIONAL SUMMARY

Experienced and results-driven Solutions Architect and Senior Engineer with over 10 years in large-scale distributed systems, real-time data streaming, and enterprise-grade data platforms. Proven leadership in designing end-to-end data solutions aligned to business objectives, regulatory requirements, and modern cloud architectures. Adept in stakeholder communication, platform governance, and solution delivery across cross-functional, multi-team environments.

SKILLS

- Data Architecture & Engineering: Large-Scale Distributed Systems, Microservices, Event-Driven Architecture, Data Lakehouse, Streaming Data Pipelines, Data Modeling (Avro, JSON, Dimensional)
- Experience with Large-Scale Distributed Systems, Microservices Design, Event-Driven Architectures, Avro/JSON Data Modeling
- Java, Kafka, KStreams DSL, Oracle, MySQL, MongoDB, SpringBoot, Openshift, Kubernetes, Docker, ELK, NodeJS, Groovy, Spock, Jenkins (CI/CD), Git

EXPERIENCE

SOLUTIONS ARCHITECT | IBM CANADA LTD.

April 2022 - Present | Vancouver, BC

- Lead application and data architect for enterprise-scale data streaming platform using Confluent Kafka and Redpanda, overseeing platform governance for 20+ development teams
- Advised on architecture best practices across ingestion, transformation, and data movement using Kafka, KStreams, and Spring Boot microservices
- Designed scalable ETL pipelines and data solutions supporting batch and real-time analytics use cases across regulatory, customer, and retail banking domains
- Designed and implemented a custom data reconciliation and governance framework for validating batch and streaming data consistency across Kafka-based pipelines; ensured alignment with data quality rules, regulatory requirements, and enterprise audit standards
- Delivered 10x faster recovery for stateful Kafka applications and reduced data ingestion bottlenecks using ETL Infortimca, enhancing throughput and resiliency of high-volume data pipelines
- Presented architecture proposals to internal review boards and stakeholder groups using detailed documentation and visual models
- Mentored junior architects and developers, fostering an engineering culture focused on excellence, reusability, and continuous improvement

SOFTWARE ENGINEER | IBM CANADA LTD. - BACKEND/DATA AND TECHNICAL PROGRAM MANAGER

April 2019 - March 2022 | Toronto, ON

- Reduced client's real-time transaction processing latency to less than 200ms using KStreams join operations with KTables and built RESTful KTable API for fast querying
- Designed and maintained LowCashMode streaming pipelines - saving customers approximately \$125 to \$150 million annually in overdraft fees with real-time overdraft alerting serving 20 million+ customer
- Supported and managed various data streaming pipelines to support increased work volumes resulting from an acquisition, allowing client to successfully finish the merger without adding staff
- Developed client's core data streaming platform utilize Confluent Kafka creating a 80 - 90% reduction in the number of loan applications that the bank must review manually
- Created a highly used JIRA dashboards for sprint planning, PowerPoint road-maps for yearly, quarterly and monthly milestones, Resource Charts for project planning and Confluence documentations for project details and technical guidance for other teams
- Wrote over 100+ Springboot microservices interfacing with Kafka via KStreams DSL and create various RESTful APIs

COMPUTER SCIENCE LECTURER | ICALC SCHOOL - PYTHON NOVICE/INTERMEDIATE

Sept 2017 - Dec 2019 | Toronto, ON

- Computer Science Teacher at ICALC, responsible for instructing elementary to middle school students in computer science fundamentals using Python. Developed comprehensive presentations, coursework, practice problems, and test materials while effectively managing a class of approximately 15 students

EDUCATION

ELECTRICAL AND COMPUTER ENGINEERING

April 2019 | University of Waterloo

- **Relevant Courses:** Data Structures & Algorithms, Computer & Communication Networks, Cooperative & Adaptive Algorithms, Operating Systems, Cryptography & System Security, Programming for Performance